HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed bit-accurate functional C++ models for new GPU architectures, closely collaborating with architects and design teams to ensure fidelity.
- Constructed comprehensive test suites for C++ models and RTL simulations, executing conformance, stress, and random tests to stabilize GPU compute systems.
- Implemented and debugged C++ code for complex architectural features, adhering to clean coding practices for maintainability and team collaboration.
- Integrated Python scripts for automating test execution and data analysis of simulation results, improving iteration speed for the core development team.
- Participated in Agile ceremonies including sprint planning and daily stand-ups, contributing to technical estimations and task breakdown for GPU core development.
- Gained exposure to graphics pipelines through collaboration with driver and compiler teams, understanding their integration points with hardware models.
- Containerized development environments and model simulation tools using Docker, facilitating consistent builds and deployments across distributed team members.

SKILLS

Languages – C++, Python

Architecture – Computer Architecture, Graphics Pipelines, GPU Cores

Development – C++ Development, Debugging, Problem-Solving, Agile Development, Scripting

APIs – DirectX, OpenGL, Vulkan, OpenCL

Tools – RTL Simulations, Docker, Git, Jira, Confluence

PROJECTS

C++ GPU Architecture Modelling Framework

- Designed and implemented a modular C++ framework for creating bit-accurate functional models of new GPU architectures, enabling early-stage design verification.
- Integrated Python scripting for automated test case generation and result aggregation, reducing manual effort by approximately 40% per simulation cycle.
- Developed reusable C++ components for common architectural elements, improving model development efficiency and consistency across multiple projects.

Accelerated GPU Simulation Test Harness

- Built a Python-based test harness to orchestrate complex simulation runs and analyze performance characteristics of GPU models.
- Implemented random stimulus generation techniques in C++ to stress-test architectural edge cases, uncovering 15 critical bugs before RTL integration.
- Collaborated with a distributed team via Confluence and Git for code reviews and feature integration, ensuring timely delivery of simulation updates.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)